

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Mathematics

MT-333

Advanced Calculus & Fourier Analysis

15-Week Course Plan

Program: BE Industrial Manufacturing Engineering

Level: Third Year Undergraduate

Duration: 15 Weeks

Course Type: Advanced Engineering Mathematics

Academic Year: 2026

1. Course Description

MT-333, *Advanced Calculus & Fourier Analysis*, develops the mathematical tools required for modeling and analyzing engineering systems involving spatial and temporal variation.

The course builds upon the concepts of differential and integral calculus and extends them to functions of several variables, partial differential equations, Fourier series, Fourier transforms, and vector calculus.

Particular emphasis is placed on engineering applications relevant to Industrial Manufacturing Engineering, including heat transfer, vibration, wave propagation, stress analysis, fluid flow, manufacturing processes, signal analysis, and field-based engineering models.

2. Course Philosophy

The course follows the mathematical progression

Multivariable Calculus $\longrightarrow$ Partial Differential Equations

Periodic Phenomena $\longrightarrow$ Fourier Analysis $\longrightarrow$ FFT and Spectrum

Vector Fields $\longrightarrow$ Vector Differentiation $\longrightarrow$ Vector Integration

Mathematical Model $\longrightarrow$ Engineering Problem $\longrightarrow$ Mathematical Solution $\longrightarrow$ Physical Interpretation

The emphasis is therefore not only on solving equations but also on understanding what the mathematical solution means in an engineering context.

3. Course Learning Objectives

By the end of the course, students should be able to:

- LO1:** Formulate basic engineering models using partial differential equations.
- LO2:** Classify and solve selected first-order and higher-order partial differential equations.
- LO3:** Solve standard wave equations using D'Alembert's method and related techniques.
- LO4:** Apply Lagrange's method to appropriate first-order partial differential equations.
- LO5:** Represent periodic functions using Fourier series.
- LO6:** Determine Fourier coefficients for general, even, odd and half-range expansions.
- LO7:** Explain the relationship between Fourier series, Fourier transforms, FFT and Fourier spectra.
- LO8:** Analyze stationary points, local extrema and saddle points of functions of several variables.
- LO9:** Evaluate double and triple integrals and apply them to engineering problems.

- LO10:** Calculate area, volume, centroid, moment of inertia and surface area using multiple integrals.
- LO11:** Calculate the gradient, directional derivative, divergence, curl and Laplacian of scalar and vector fields.
- LO12:** Identify conservative, solenoidal and irrotational vector fields.
- LO13:** Evaluate line and surface integrals.
- LO14:** Apply Green's theorem, Gauss' divergence theorem and Stokes' theorem to engineering problems.
- LO15:** Transform multiple integrals between coordinate systems using Jacobians.
- LO16:** Interpret advanced calculus concepts in the context of manufacturing and engineering systems.

4. 15-Week Course Schedule

Table 1: Sequential 15-Week Course Plan for MT-333

Week	Main Topic	Key Concepts / Skills	Engineering Application
1	Advanced Multi-variable Calculus	Functions of several variables; stationary points; higher partial derivatives; local maxima/minima; saddle points	Optimization of manufacturing parameters
2	Multiple Integrals	Iterated integrals; double integrals; triple integrals; change of order	Area, volume and mass distribution
3	Engineering Applications of Multiple Integrals	Centroid; center of mass; moment of inertia; surface area; volume	Mechanical components and material distribution
4	Introduction to PDEs	Formation of PDEs; basic concepts; order and degree; classification; linear and homogeneous PDEs	Modeling heat, vibration and manufacturing processes
5	First-Order PDEs	First-order linear PDEs; special forms; Lagrange's solution	Transport and manufacturing process models
6	Higher-Order PDEs	Linear homogeneous PDEs; auxiliary equations; relationship with ODEs; standard forms	Engineering system models
7	Wave Equation	D'Alembert's solution; one-dimensional wave equation; two-dimensional wave equation	Vibration, machining dynamics and wave propagation
8	Fourier Series	Periodic functions; Fourier coefficients; expansion with arbitrary periods	Periodic manufacturing signals and vibrations

Week	Main Topic	Key Concepts / Skills	Engineering Application
9	Fourier Series: Symmetry and Half-Range	Even and odd functions; half-range sine and cosine series; convergence	Boundary-value problems and heat transfer
10	Fourier Transform, FFT and Spectrum	Fourier transform concept; discrete Fourier transform; FFT; frequency-domain representation; Fourier spectrum	Machine vibration and signal analysis
11	Vector Differential Calculus	Vector differential operator; gradient; directional derivative; divergence; curl; Laplacian	Temperature, pressure and velocity fields
12	Vector Fields	Conservative and solenoidal fields; irrotational fields; potential functions	Fluid flow and force fields
13	Line and Surface Integrals	Line integrals; work done; surface integrals; Jacobian and coordinate transformation	Work, flux and field-based engineering models
14	Integral Theorems	Green's theorem; Gauss' divergence theorem; Stokes' theorem	Flux, circulation and conservation laws
15	Integrated Engineering Applications	Review and integration of PDEs, Fourier analysis, multiple integrals and vector calculus	Manufacturing vibration, heat transfer and engineering field models

5. Detailed Weekly Plan

5.1. Week 1 – Advanced Multivariable Calculus

Topics

- Functions of several variables.
- Stationary points.
- Local maximum and minimum.
- Saddle points.
- Critical points of functions of two variables.
- Higher partial derivatives.
- Second derivative test.

For

$$z = f(x, y),$$

a stationary point satisfies

$$\frac{\partial f}{\partial x} = 0, \quad \frac{\partial f}{\partial y} = 0.$$

Introduce the Hessian determinant

$$D = f_{xx}f_{yy} - (f_{xy})^2.$$

The classification of stationary points should be connected to engineering optimization.

Manufacturing Application

Optimization of machining parameters such as

cutting speed, feed rate, depth of cut.

5.2. Week 2 – Multiple Integrals

Topics

- Iterated integrals.
- Double integrals.
- Reversing the order of integration.
- Triple integrals.
- Rectangular and non-rectangular regions.
- Coordinate transformations.

Introduce

$$\iint_R f(x, y) \, dA$$

and

$$\iiint_V f(x, y, z) \, dV.$$

Engineering Interpretation

Multiple integration can be used to determine total quantities distributed over an area or volume.

5.3. Week 3 – Engineering Applications of Multiple Integrals

Topics

- Area.
- Volume.

- Mass.
- Centroid.
- Center of mass.
- Moment of inertia.
- Surface area.

For example,

$$M = \iint_R \rho(x, y) dA$$

represents mass for a lamina with density $\rho(x, y)$.

Centroid coordinates can be obtained from

$$\bar{x} = \frac{1}{M} \iint_R x \rho(x, y) dA,$$

$$\bar{y} = \frac{1}{M} \iint_R y \rho(x, y) dA.$$

Industrial Manufacturing Applications

- Material distribution.
- Center of mass of components.
- Rotational inertia.
- Mechanical design.
- Manufacturing component analysis.

5.4. Week 4 – Introduction to Partial Differential Equations

Topics

- Definition of a partial differential equation.
- Formation of PDEs.
- Order and degree.
- Linear and nonlinear PDEs.
- Homogeneous PDEs.
- Classification.
- Relation between PDEs and ODEs.

A general PDE may be represented as

$$F(x, y, u, u_x, u_y, u_{xx}, \dots) = 0.$$

Explain the difference between

$$\frac{dy}{dx}$$

and

$$\frac{\partial u}{\partial x}.$$

Engineering Motivation

PDEs arise naturally in:

Heat Transfer Vibration Fluid Flow Stress Analysis

5.5. Week 5 – First-Order PDEs

Topics

- First-order PDEs.
- Linear first-order PDEs.
- Special types.
- Lagrange's method.
- Complete and general integrals.

For a PDE of the form

$$Pp + Qq = R,$$

where

$$p = \frac{\partial z}{\partial x}, \quad q = \frac{\partial z}{\partial y},$$

introduce Lagrange's auxiliary equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}.$$

5.6. Week 6 – Higher-Order PDEs

Topics

- Linear homogeneous PDEs.
- Higher-order PDEs.
- Auxiliary equations.
- Complementary function.
- Particular integral.
- Relationship between PDEs and ODEs.
- Standard forms.

For example,

$$F(D, D')z = 0,$$

where

$$D = \frac{\partial}{\partial x}, \quad D' = \frac{\partial}{\partial y}.$$

The connection with ordinary differential equations should be emphasized.

5.7. Week 7 – Wave Equation

Introduce the one-dimensional wave equation

$$\boxed{\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}.$$

Introduce D'Alembert's solution

$$u(x, t) = F(x - ct) + G(x + ct).$$

Discuss:

- Wave propagation.
- Boundary and initial conditions.
- One-dimensional waves.
- Two-dimensional wave equation.

Industrial Manufacturing Applications

- Machine-tool vibration.

- Structural vibration.
- Mechanical waves.
- Noise and vibration analysis.
- Dynamic response of manufacturing systems.

5.8. Week 8 – Fourier Series

Topics

- Periodic functions.
- Fourier series.
- Fourier coefficients.
- Arbitrary periods.
- Convergence of Fourier series.

For a function of period $2L$,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right].$$

The coefficients are

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx,$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

5.9. Week 9 – Symmetry and Half-Range Fourier Series

Topics

- Even functions.
- Odd functions.
- Fourier series of even functions.
- Fourier series of odd functions.
- Half-range sine series.
- Half-range cosine series.

For an even function,

$$b_n = 0.$$

For an odd function,

$$a_n = 0.$$

This substantially simplifies Fourier calculations.

Application

Half-range expansions are particularly useful in boundary-value problems involving heat transfer and vibration.

5.10. Week 10 – Fourier Transform, FFT and Fourier Spectrum

Topics

- Fourier transform concept.
- Frequency-domain representation.
- Discrete Fourier Transform.
- Fast Fourier Transform.
- Fourier spectrum.
- Amplitude and frequency.

Introduce the Fourier transform conceptually:

Time Domain $\longleftrightarrow$ Frequency Domain
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Industrial Manufacturing Application

A vibration signal measured from a rotating machine can be transformed into a frequency spectrum. Students can investigate how a machine's vibration signal can reveal possible faults.

Suggested Practical

Record or provide a vibration signal and ask students to identify dominant frequencies using FFT.

5.11. Week 11 – Vector Differential Calculus

Introduce the vector differential operator

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}.$$

Topics

- Gradient.

- Directional derivative.
- Divergence.
- Curl.
- Laplacian.

For a scalar field f ,

$$\nabla f$$

is the gradient.

The directional derivative in the direction of a unit vector $\mathbf{u}$ is

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}.$$

For a vector field $\mathbf{F}$,

$$\nabla \cdot \mathbf{F}$$

is the divergence, while

$$\nabla \times \mathbf{F}$$

is the curl.

The Laplacian is

$$\nabla^2 f = \nabla \cdot \nabla f.$$

5.12. Week 12 – Vector Fields

Topics

- Scalar and vector fields.
- Conservative vector fields.
- Potential functions.
- Solenoidal vector fields.
- Irrotational fields.

A vector field $\mathbf{F}$ is conservative if

$$\mathbf{F} = \nabla \phi$$

for some scalar potential ϕ .

A conservative field satisfies, under appropriate conditions,

$$\nabla \times \mathbf{F} = \mathbf{0}.$$

A solenoidal field satisfies

$$\nabla \cdot \mathbf{F} = 0.$$

Applications

- Fluid velocity fields.
- Heat flow.
- Force fields.
- Electromagnetic fields.
- Pressure fields.

5.13. Week 13 – Line and Surface Integrals

Topics

- Parametric curves.
- Line integrals.
- Work done by a vector field.
- Surface integrals.
- Flux.
- Jacobian transformation.

A line integral of a vector field is

$$\int_C \mathbf{F} \cdot d\mathbf{r}.$$

For a force field, this represents work:

$$W = \int_C \mathbf{F} \cdot d\mathbf{r}.$$

Surface flux can be written as

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS.$$

5.14. Week 14 – Green's, Gauss' and Stokes' Theorems

This week should be treated as a major synthesis of vector calculus.

Green's Theorem

$$\oint_C P dx + Q dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

Gauss' Divergence Theorem

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_V \nabla \cdot \mathbf{F} dV.$$

Stokes' Theorem

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS.$$

Emphasize the common idea:

$$\text{Boundary Information} \longleftrightarrow \text{Interior Information}$$

5.15. Week 15 – Integrated Engineering Applications

The final week should not introduce a large new mathematical topic.

Instead, integrate the semester's major ideas through engineering problems.

Possible case studies include:

1. Vibration analysis of a manufacturing machine using Fourier analysis.
2. Heat distribution using a partial differential equation.
3. Material distribution and moment of inertia using multiple integrals.
4. Fluid flow using divergence and curl.
5. Work and circulation using line integrals.
6. Flux through an engineering surface using Gauss' theorem.
7. Boundary circulation using Stokes' theorem.

6. Mathematical Narrative of the Course

The course can be presented as a continuous mathematical story:

Multivariable Calculus

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Multiple Integrals



Engineering Quantities



Partial Differential Equations



Wave and Heat Models



Fourier Series



Fourier Transform / FFT



Vector Fields



Line and Surface Integrals



Green + Gauss + Stokes



Engineering Modeling

7. Industrial Manufacturing Applications

The following applications should be repeatedly incorporated into lectures, tutorials, and assignments.

Mathematical Topic	Industrial Manufacturing Application
Multivariable Optimization	Machining parameter optimization; production efficiency
Multiple Integrals	Mass, centroid, volume and moment of inertia
Partial Differential Equations	Heat transfer, vibration and material processes
Wave Equation	Machine vibration and structural dynamics
Fourier Series	Periodic machine signals and vibration patterns
FFT	Machine fault detection and predictive maintenance
Gradient	Temperature and pressure variation
Divergence	Flow and source/sink behavior
Curl	Rotation and circulation of fluid fields
Line Integrals	Work performed by forces along manufacturing paths
Surface Integrals	Flux through engineering surfaces
Gauss' Theorem	Conservation and flux calculations
Stokes' Theorem	Circulation and rotational field analysis

8. Suggested Practical / Computational Activities

To make the course more relevant to Industrial Manufacturing Engineering, the following activities are recommended.

1. Machining Optimization

Given a simplified model

$$Q = Q(v, f, d),$$

where v is cutting speed, f is feed rate and d is depth of cut, students determine optimal operating conditions.

2. Heat Distribution

Solve or investigate a simplified heat equation

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}.$$

3. Machine Vibration

Generate a periodic vibration signal and represent it using a Fourier series.

4. FFT Demonstration

Use a computational tool to transform a vibration signal from the time domain to the frequency domain.

5. Centroid and Moment of Inertia

Use double integrals to determine geometric properties of an irregular manufacturing component.

6. Vector Field Visualization

Plot a vector field and investigate its gradient, divergence and curl.

7. Work Along a Manufacturing Path

Calculate

$$W = \int_C \mathbf{F} \cdot d\mathbf{r}$$

for a force acting along a specified path.

8. Flux Calculation

Use a surface integral or Gauss' theorem to calculate flux through an engineering surface.

9. Suggested Weekly Teaching Structure

A typical week may be organized as follows:

Session	Purpose
Lecture 1	Conceptual introduction and mathematical foundations.
Lecture 2	Derivation, mathematical techniques and worked examples.
Lecture 3	Engineering applications and problem solving.
Tutorial / Lab	Computational activity, numerical experiment or problem-solving session.

10. Recommended Assessment Structure

A possible assessment distribution is:

Assessment Component	Suggested Weight
Quizzes	10%
Assignments / Class Work	10%
Computational / Practical Activities	10%
Midterm Examination	25%
Final Examination	40%
Engineering Mathematics Project	5%
Total	100%

11. Recommended Project Themes

For a third-year Industrial Manufacturing Engineering class, students can work in small groups on one of the following:

1. Machine Vibration Analysis Using FFT
2. Heat Distribution in a Manufacturing Component
3. Optimization of Machining Parameters
4. Centroid and Moment of Inertia of an Irregular Component
5. Fourier Analysis of a Periodic Manufacturing Signal
6. Vector Field Modeling of Fluid Flow
7. Work and Energy Along a Manufacturing Path
8. Flux Analysis of an Engineering Surface

12. Recommended Books

The following references are recommended for the course.

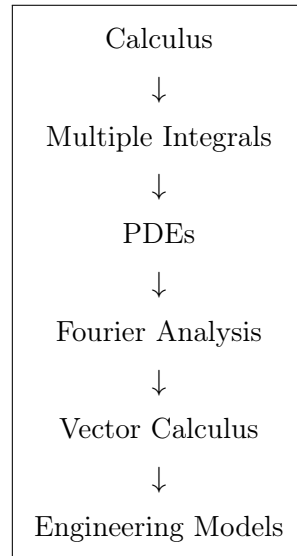
1. Erwin Kreyszig, *Advanced Engineering Mathematics*, 7th Edition.
2. Howard Anton, *Calculus and Analytical Geometry*, 5th Edition.
3. J. Farlow, *Introduction to Differential Equations*, 1994.
4. G. Zill, *Differential Equations*.

Additional references may be used for Fourier analysis, vector calculus, and engineering applications.

13. Final Course Perspective

The central objective of MT-333 is to move students from computational calculus toward mathematical modeling of real engineering systems.

The progression can be summarized as



The student should ultimately understand that

Engineering Problem $\rightarrow$ Mathematical Model $\rightarrow$ Mathematical Analysis $\rightarrow$ Numerical/Computational Solution

Advanced Mathematics is a language for modeling engineering systems.

From differential equations and Fourier analysis to manufacturing, vibration, heat transfer and field problems.